POULTRYWISE KNOWLEDGE SERIES

Book 5: Poultry Feed Milling & Nutrition Masterclass

A Comprehensive Guide to On-Farm Feed Formulation, Ingredient Quality & Mill Processing

PoultryWise Operational Manuals • Technical Milling & Formulations Standard

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CHAPTER 1: RAW MATERIAL SELECTION & QUALITY ASSURANCE

Feed represents 65% to 75% of total poultry production costs. Achieving maximum bio-economic performance relies on strict raw material inspection and quality control before processing.

- **Moisture Content Limits:** Grains (maize, wheat) must remain below 12.5% to 13.0% moisture to prevent mold growth and mycotoxin formation.
- **Mycotoxin Risk Management:** Screen incoming corn and soybean meal batches for Aflatoxin (< 20 ppb) and Ochratoxin A. Utilize broad-spectrum toxin binders in feed during high-humidity seasons.
- **Adulteration Testing:** Check protein meals (soybean meal, fishmeal) for synthetic urea contamination or high ash content indicating sand adulteration.

CHAPTER 2: ENERGY & PROTEIN CHARACTERIZATION

Poultry eat primarily to meet their metabolizable energy (ME) requirements. Feed formulation requires balancing available energy with digestible amino acid profiles.

Raw Material	Crude Protein (CP %)	Metabolizable Energy (kcal/kg)	Key Inclusion Constraints
Yellow Corn / Maize	8.5%	3,350	Base energy source (up to 65% inclusion)
Soybean Meal (44-48%)	46.0%	2,230	Primary protein source; check urease activity
Wheat Bran / Rice Bran	12.5% – 14.0%	1,800 – 2,000	Fiber source; max 8% in broilers, 12% in layers
Vegetable Oil / Fat	0.0%	8,800	Energy booster; max 3% to 5% inclusion

CHAPTER 3: MINERAL & VITAMIN NUTRITION

Proper skeletal development, eggshell formation, and physiological enzyme pathways depend on precise micro-nutrient balance.

- **Calcium to Phosphorus Ratio:** Broilers require a 2:1 Calcium to Available Phosphorus ratio. Layer feeds require up to 10:1 in peak production to supply 4.0g Calcium per egg.
- **Coarse vs. Fine Limestone:** Layer feeds should include 60%–70% coarse limestone (2–4mm) for nighttime calcium release in the gizzard during shell formation.
- **Electrolyte Balance (dEB):** Maintain dietary electrolyte balance (Na + K - Cl) between 240 and 260 mEq/kg to prevent heat stress acidosis.

CHAPTER 4: LEAST-COST FEED FORMULATION PRINCIPLES

Least-Cost Formulation (LCF) software uses linear programming to balance target nutrient requirements against fluctuating market prices of raw materials.

Core Principles of Digestible Amino Acid Formulation

Formulate feeds based on **Digestible Amino Acids (dAA)** rather than total crude protein. Standardizing diets relative to Digestible Lysine (Ideal Protein Concept) ensures optimized muscle synthesis and minimal nitrogen excretion:

- **Digestible Methionine + Cysteine:** 72% to 75% of Lysine ratio
- **Digestible Threonine:** 63% to 65% of Lysine ratio
- **Digestible Tryptophan:** 16% to 18% of Lysine ratio

CHAPTER 5: STAGE-SPECIFIC FEED SPECIFICATIONS

Nutritional needs vary substantially across the bird's lifespan, requiring precise formulation adjustments across production phases.

Diet Phase	Target ME (kcal/kg)	Crude Protein (%)	Dig. Lysine (%)	Calcium (%)
Broiler Starter (Days 0–10)	3,000	22.5%	1.28%	0.96%
Broiler Grower (Days 11–24)	3,100	20.5%	1.15%	0.88%
Broiler Finisher (Days 25–Market)	3,200	19.0%	1.02%	0.78%
Layer Production (Peak Lay)	2,750	17.5%	0.88%	4.10%

CHAPTER 6: FEED ADDITIVES, ENZYMES & PREMIXES

Non-nutritive feed additives boost nutrient digestability, intestinal health, and operational performance.

- **Exogenous Enzymes:** Phytase releases bound phytate-phosphorus, reducing inorganic phosphate costs. Xylanase/Glucanase breaks down non-starch polysaccharides (NSP) in wheat and barley.
- **Gut Health Conditioners:** Organic acids (formic/propionic), probiotics, and essential oils lower gut pH and inhibit *Salmonella* and *Clostridium perfringens*.
- **Antioxidants & Mold Inhibitors:** Protect fats against rancidity and prevent storage mold development.

CHAPTER 7: FEED PROCESSING: GRINDING & PARTICLE SIZE

Grinding breaks down raw grains to increase surface area for digestive enzyme activity and ensure uniform mixing.

- **Hammer Mill Setup:** Screen size determines particle distribution. Use 3.0mm to 3.5mm screens for broilers and 4.0mm to 5.0mm screens for laying hens.
- **Target Geometric Mean Diameter (d50):**
  - Broilers (Starter): 700 – 900 microns
  - Broilers (Finisher): 900 – 1,100 microns
  - Layer Mash: 1,200 – 1,500 microns (coarser feed promotes gizzard development)

CHAPTER 8: MIXING DYNAMICS & HOMOGENEITY

Mixing is the most critical physical process in feed manufacturing. Micro-ingredients (premixes, amino acids) must be uniformly distributed throughout every batch.

Mixer Performance & Coefficient of Variation (CV)

Mixing quality is measured using a **Coefficient of Variation (CV)** test with salt or synthetic tracer analysis:

- **CV < 5%:** Excellent mixing efficiency
- **CV 5% – 10%:** Acceptable standard operational performance
- **CV > 10%:** Poor uniformity — leads to bird performance variation; requires mixer ribbon/paddle adjustment or extended mixing cycle time.

CHAPTER 9: CONDITIONING, PELLETING & CRUMBLING

Pelleting thermally processes feed to improve digestibility, eliminate pathogens, and reduce feed waste.

- **Steam Conditioning:** Expose feed to $80^\circ ext{C} - 85^\circ ext{C}$ steam for 30–45 seconds to gelatinize starch and improve pellet durability.
- **Pellet Durability Index (PDI):** Measure using a Holmen or Tumbler tester. Target a PDI above 90% for finisher broilers.
- **Crumbling:** Pass pellets through corrugated roller mills to produce crumble diets tailored for young chicks (Days 0–10).

CHAPTER 10: QUALITY CONTROL & STORAGE MANAGEMENT

Maintaining finished feed quality prevents nutrient degradation and biological contamination prior to consumption.

- **Bin & Silo Management:** Enforce First-In, First-Out (FIFO) stock rotation. Clean feed bins every cycle to remove caked feed buildup.
- **Finished Feed Shelf Life:** Mash feeds with added fats should be used within 14 days. Pelleted feeds stored under cool conditions remain stable for up to 30 days.
- **Batch Traceability:** Retain feed samples (500g) from every manufactured batch for 60 days to facilitate diagnostic testing during flock performance inquiries.

End of Book 5 — PoultryWise Knowledge Series • Feed Milling & Nutrition Masterclass